

Problem Set 6

Throughout this problem set, use Black's formula to price up options and assume flat skew (i.e., the level of vol used to price a swaption does not vary with strike) unless explicitly stated otherwise.

For questions 1–4 and 7, suppose that the swap curve is as follows:

Maturity	Swap Rate
1 year	6.00%
2 year	6.25%
3 year	6.50%
4 year	6.75%
5 year	7.00%

Assume that these rates are annual bond versus 1-year LIBOR annual bond.

For questions 1–4 and 7, when pricing swaptions using Black's formula, use the lognormal swaption vol matrix in Table 5.1 to determine the appropriate mid-market volatility. Unless stated otherwise, all pricing is done at mid-market.

1. Answer the following questions:

a) What is the 5-year swap rate?

b) Consider a 5nc2 swap, a 5-year swap in which a swap dealer pays fixed to a counterparty for five years and owns the right to terminate in two years (single European). What type of swaption is embedded into this cancelable swap? Who is long and who is short the swaption? What is the 5nc2 swap rate?

c) Consider a 5-year cancelable swap in which a swap dealer pays fixed to a counterparty, and the counterparty pays 1-year LIBOR - 25 bps. The swap dealer owns the right to terminate in two years (single European). What is the fixed rate in this cancelable swap?

d) A *puttable swap* is a swap in which one counterparty receives fixed and owns the right to terminate prior to the stated maturity. Consider a 5np2

(5 nonput 2), a 5-year swap in which a swap dealer receives fixed for five years and the swap dealer owns the right to terminate the swap after two years (single European). What type of swaption is embedded into this swap? Who is long and who is short? What is the 5np2 swap rate?

e) Consider a 5-year swap in which a swap dealer pays fixed versus 1-year LIBOR and the swap dealer has the right to terminate the swap after two years. The counterparty has specified that it would like to be paid a fixed coupon equal to 7.75% in the first two years. What fixed coupon can be paid in the back three years so that the trade has zero value today?

2. Suppose that one year ago we entered into a 4nc1 Bermudan cancelable swap, with particular cancelation dates. When we entered into the swap, we agreed to pay fixed for four years with the right to cancel in one year and in two years. The fixed rate in this 4nc1 Bermudan is 6.75%.

Suppose now a year has gone by and we are at the first cancelation date. Should we cancel the swap?

3. Suppose that one year ago we entered into a 3nc1 Bermudan cancelable swap. When we entered into the swap originally, we agreed to pay fixed for three years with the right to cancel annually. The fixed rate in this trade is 6.77%. Suppose we are now at the first cancelation date.

a) As it stands here today, what option position is embedded in the remaining trade? Given this option position, what lognormal implied vol should be used to value the residual trade assuming flat skew (i.e. assuming vol does not vary with strike)?

b) Given the lognormal implied vol for swaptions struck at-the-money forward presented in Table 5.1, what is the equivalent normal implied vol for this option assuming an at-the-money forward strike?

c) Under the assumption of flat lognormal skew (i.e., lognormal implied vol does not vary with strike), should we cancel the swap if we employ Black's model?

d) Under the assumption of flat normal skew (i.e., normal implied vol does not vary with strike), should we cancel the swap if we employ the normal model?

e) Are your answers to c) and d) the same? Explain.

4. Consider a 4nc1, a 4-year swap in which a dealer pays fixed and has the one-time right to cancel after one year.

a) What type of option (identified by its expiry and tail) is embedded into this trade? For this particular option, what is the mid-market level of log-normal implied vol for the at-the-money forward strike, using the lognormal swaption vol matrix in Table 5.1?

b) Given your answer to part a), what is the level of normal implied vol for the at-the-money forward strike of this option?

c) Assume that the dealer prices up the 4nc1 using a lognormal model (i.e., using Black's formula) and makes the assumption of flat lognormal skew. What is the 4nc1 swap rate?

d) Assume that the dealer prices up the 4nc1 using a normal model and makes the assumption of flat normal skew. What is the 4nc1 swap rate?

e) How does the rate you came up with in part c) compare to the rate you came up with in part d)? Explain.

5. Consider again our stylized example in which swaps pay annual bond on the fixed side versus 1-year LIBOR. Suppose that a dealer and a client enter into a 5nc1 Bermudan cancelable swap. In this trade, the dealer pays a fixed rate of 6% for five years and has the right to cancel annually starting in one year.

Suppose now that one year has gone by. Suppose that the par rate on a 4nc1 Bermudan cancelable swap (with the right to cancel annually) is currently 5.50%.

The dealer currently has the right to cancel the swap that it entered into a year ago. The dealer is concerned that if it does cancel the swap today, the client may enter into another swap and possibly choose a different dealer with which to execute this new trade. Suppose the dealer approaches the client and says that instead of canceling the trade, the dealer will keep the trade outstanding if the client is willing to move the coupon down from 6% to 5.75%.

Is this a good proposal for the dealer to be making?

6. Consider an index amortizing swap that has a 5-year final maturity and a 1-year lockout. The notional on the swap may amortize in one year in accordance with the following schedule:

Index Amortization Schedule	
3-month LIBOR	Percentage of Remaining Notional that Amortizes
$\leq L_L$	100%
$= L_M$	50%
$\geq L_H$	0%

Here $L_L < L_M < L_H$. Linear interpolation is used to determine the percentage of amortization that will occur for levels of LIBOR that are not explicitly shown on the schedule. For example, if 3-month LIBOR were equal to $(L_M + L_H)/2$, then 25% of the remaining notional would amortize at that time.

a) Assume that the index amortizing swap is “single European” in nature, meaning that amortization of the notional can occur only at one time. Assume that this occurs at the end of year one. That is, we will look at the schedule one time only at the end of the first year. Whatever the notional is after one year will be the notional for the remaining four years of the trade. Denote the fixed coupon in this trade by f_i and assume without loss of generality that this coupon is paid quarterly 30/360 versus 3-month LIBOR.

Now consider a single European 5nc1, a 5-year swap in which one counterparty pays fixed and has the one-time right to cancel in one year. Assume that the fixed coupon in this trade is also paid quarterly 30/360 versus 3-month LIBOR and denote the fixed coupon by f_c .

Assuming that all transactions are done at mid-market, show that $f_i \leq f_c$.

(*Hint:* You can show that this is true by setting up an arbitrage argument. Assume that $f_i > f_c$ and show that this assumption will lead to arbitrage. In doing so, you will have to consider all cases of what may happen in the future.)

b) Assume that the index amortizing swap is “multi-European” or “Bermudan” in nature. In particular, amortization of the notional can occur in one year and in two years. That is, we will look at the schedule at the end of the first year to determine the notional for the second year. And we will look at the schedule at the end of the second year to determine the notional for the remaining three years of the trade. Denote the fixed coupon in this trade by F_i and assume without loss of generality that this coupon is paid quarterly 30/360 versus 3-month LIBOR.

Now consider a multi-European or Bermudan 5nc1, a 5-year swap in which one counterparty pays fixed and has the right to cancel in one year and, if not canceled at the end of the first year, the counterparty has the right to cancel the trade at the end of the second year. Assume that the fixed coupon in this trade is also paid quarterly 30/360 versus 3-month LIBOR and denote the fixed coupon by F_c .

Assuming that all transactions are done at mid-market, show that $F_i \leq F_c$.

(*Hint:* You will also need to set up an arbitrage argument here as well. Assume that $F_i > F_c$ and show that this assumption will lead to arbitrage. In doing so, you will have to consider all cases of what may happen in the future. You can save yourself quite a bit of time on part b) if you generalize your result from part a); that is, if you recognize that it applies to any single European index and cancelable swap with the same final maturity and lockout/cancellation date, then apply it appropriately to particular parts of part b).)

7. Consider an *extendible swap* in which a dealer pays fixed for one year and receives LIBOR and has the right to extend the trade for an additional three years at the same fixed rate. That is, at the end of the first year, the dealer may elect to continue paying the same fixed rate and receive LIBOR for an additional three years. If the dealer does not elect to extend the trade, then coupon payments will be made at the end of the first year, and the trade will end.

a) What is the fixed rate in this trade?

b) How does the fixed rate in this extendible swap compare to the fixed rate in a 4nc1 that you solved for in problem 4 c)? Explain.

8. Suppose that today’s swap curve is as follows (these rates annual bond and assume that the floating side is 1-year LIBOR paid annual bond as well):

Maturity	Swap Rate
1 year	5.25%
2 year	5.50%
3 year	5.75%
4 year	6.00%
5 year	6.25%

A (*period-by-period*) *knockout swap* is a fixed-floating swap in which payments on each side can be “knocked out” if LIBOR rates reach a certain level. To take a concrete example, consider a \$100mm 5-year swap in which a client receives fixed annual bond and pays 1-year LIBOR to the swap dealer Swaps 'R Us.

This swap has the following knockout feature: if, on any LIBOR reset date, 1-year LIBOR is less than or equal to 5.00%, then neither the fixed nor the floating payment is made for that period. If payments are knocked out in any one period, they can certainly resume the next period, and will, provided that LIBOR is above 5.00%.

To summarize, the structure of the swap is as follows:

Swaps 'R Us Pays:	Fixed
Client Pays:	1-year LIBOR
Notional:	\$100mm
Maturity:	5 years
Payment Frequency:	Annual, on both sides
Day Count:	30/360, on both sides
Knockout Feature:	If, on any reset date, 1-year LIBOR is less than or equal to 5.00%, then there will be no payments on either side for that period

Furthermore, assume that the base vol curve is flat at 20%.

a) What is the fixed coupon in the swap? That is, what fixed coupon gives this swap an $NPV = 0$?

b) What is the value of a *vega* in this trade, defined to be the change in the value of the trade, given a 1% parallel shift upwards in the base vol curve? That is, once the fixed coupon has been set, if each of the base vols is increased 1%, how does this affect the value of the trade? Which counterparty is long vol in this trade?

c) Suppose now that one year has gone by. Assume that the swap curve is as follows:

Maturity	Swap Rate
1 year	4.75%
2 year	5.00%
3 year	5.25%
4 year	5.50%
5 year	5.75%

Furthermore, assume that the base vol curve is flat at 20%. What is the mark-to-market of the trade? Whose favor is the trade in?

For questions 9–11 below, suppose that the swap curve today is as follows (these rates are annual bond and assume that the floating side is 1-year LIBOR paid annual bond as well):

Maturity	Swap Rate
1 year	5.25%
2 year	5.50%
3 year	5.75%
4 year	6.00%
5 year	6.25%

In addition, suppose that today's base vol curve is as follows:

Forward Period	Base Vol
0x1	n.a.
1x2	16%
2x3	xx%
3x4	15.9%
4x5	xx%

and also suppose that today's swaption matrix is as follows:

		Underlying Swap (Tail)									
		1	2	3	4	5	7	10	15	20	30
E x p i r y	1m	12.7	14.7	14.6	14.5	14.3	13.5	12.2	11.5	11.3	11.0
	3m	11.9	14.3	14.1	14.0	13.8	13.1	12.2	11.5	11.0	10.8
	6m	13.8	15.1	14.9	14.6	14.3	13.7	12.7	13.0	11.6	11.3
	1y	xx.x	15.8	xx.x	15.0	14.6	13.7	12.7	12.0	11.6	11.3
	2y	16.4	16.0	15.6	15.2	14.9	14.2	13.4	12.6	12.3	12.0
	3y	xx.x	15.6	15.3	15.0	14.8	14.2	13.4	12.6	12.3	11.9
	4y	15.5	15.3	15.0	14.8	14.5	14.0	13.3	12.5	12.2	11.8
	5y	15.4	15.0	14.8	14.5	14.3	13.8	13.1	12.4	12.0	11.7
	7y	14.4	14.1	13.9	13.7	13.6	13.1	12.5	11.8	11.3	11.0
	10y	13.3	13.0	12.9	12.8	12.7	12.3	11.8	11.1	10.6	10.3

In addition, for questions 9–11, suppose that the swap curve *one year from today* is as follows (these rates are annual bond and assume that the floating side is 1-year LIBOR paid annual bond as well):

Maturity	Swap Rate
1 year	5.00%
2 year	5.25%
3 year	5.50%
4 year	5.75%
5 year	6.00%

In addition, for questions 9–11, suppose that the swap curve *two years from today* is as follows (these rates are annual bond and assume that the floating side is 1-year LIBOR paid annual bond as well):

Maturity	Swap Rate
1 year	5.50%
2 year	5.75%
3 year	6.00%
4 year	6.25%
5 year	6.50%

Also, for questions 9–11, suppose that the base vol curve and swaption matrix *one year from today* are the same as they are today. Also suppose that

the base vol curve and swaption matrix *two years from today* are the same as they are today.

9. There is a relationship between the base vol curve and the swaption matrix. Answer the following questions in the paradigm of our stylized example.

- a) What is 2x3 base vol?
- b) What is 4x5 base vol?
- c) What is 1 into 1 swaption vol?
- d) What is 3 into 1 swaption vol?
- e) Can you determine what 1 into 3 swaption vol is? If you cannot determine it exactly, can you place an upper or lower bound on its value?

10. Consider a client who is interested in looking at an alternative to a standard receiver swaption, to potentially cheapen the premium needed to pay for the option.

- a) What is the price today of a 2 into 2 receiver swaption struck at 6.50%?
- b) Consider now the following alternative to the standard receiver swaption presented in part a) above. This is also a 2 into 2 receiver swaption struck at 6.50%; however, in order to be effective at expiration, the 2-year swap rate at expiration must be at or below 6.40%. In other words, if the 2-year swap rate at expiration is less than or equal to 6.40%, then the holder of the option gets the payoff from the 2 into 2 receiver swaption struck at 6.50%. But if the 2-year swap rate at expiration is above 6.40%, then there is no payoff. What is the value of this option today?
- c) If the client considers buying the option in part b) instead of the option in part a), what type of view do you think he has?
- d) Suppose that the client purchases the option in part b). Suppose now that one year goes by. What is the mark-to-market value of the option?
- e) Suppose now that two years have gone by. Today is the exercise date of the option. Instead of physically entering into the underlying swap, assume

both counterparties have agreed to cash settle the option. What is the cash settlement value of the option?

11. A *flip-flop swap* is a swap in which one counterparty has the right to switch the coupon it pays from fixed to floating after a certain period of time. As an example, consider a \$100mm 4-year swap in which Swaps 'R Us, a major swap dealer, faces one of its clients. The client pays 1-year LIBOR. Swaps 'R Us pays a coupon that may be flip-flopped. To be more specific, Swaps 'R Us will pay a fixed coupon of $x\%$. After two years, Swaps 'R Us has the right to switch the coupon for the remaining two years in the swap to 1-year LIBOR – 25 bps.

To summarize, the structure of the swap is as follows:

Swaps 'R Us Pays:	Flip-Flop Coupon
Client Pays:	1-year LIBOR
Flip-Flop Coupon:	$x\%$, subject to Flip-Flop Option
Notional:	\$100mm
Maturity:	4 years
Payment Frequency:	Annual, on both sides
Day Count:	30/360, on both sides
Flip-Flop Option:	After two years, Swaps 'R Us has the right to pay 1-year LIBOR – 25 bps for the remaining two years of the swap

Please note that the option to flip-flop the coupon after two years is “single European” in nature; if Swaps 'R Us decides to switch the coupon from fixed to floating in two years, then the final two payments made by Swaps 'R Us will be floating equal to 1-year LIBOR – 25 bps. If Swaps 'R Us decides to keep the coupons fixed after two years, then the final two coupons will be fixed coupons equal to $x\%$.

- Based on the market today, what is the fixed rate x in the swap?
- Suppose now that one year has gone by. What is the mark-to-market value of the swap that you structured in part a)?
- Suppose now that two years have gone by. Today is the day that Swaps 'R Us must decide if it wants to continue to pay fixed coupons of $x\%$ for the remaining two years of the swap or if it instead wants to pay floating coupons